MILITARY SPECIFICATION SHEET
SYNCHRO CONTROL DIFFERENTIAL TRANSMITTER, TYPE 37CDX6a

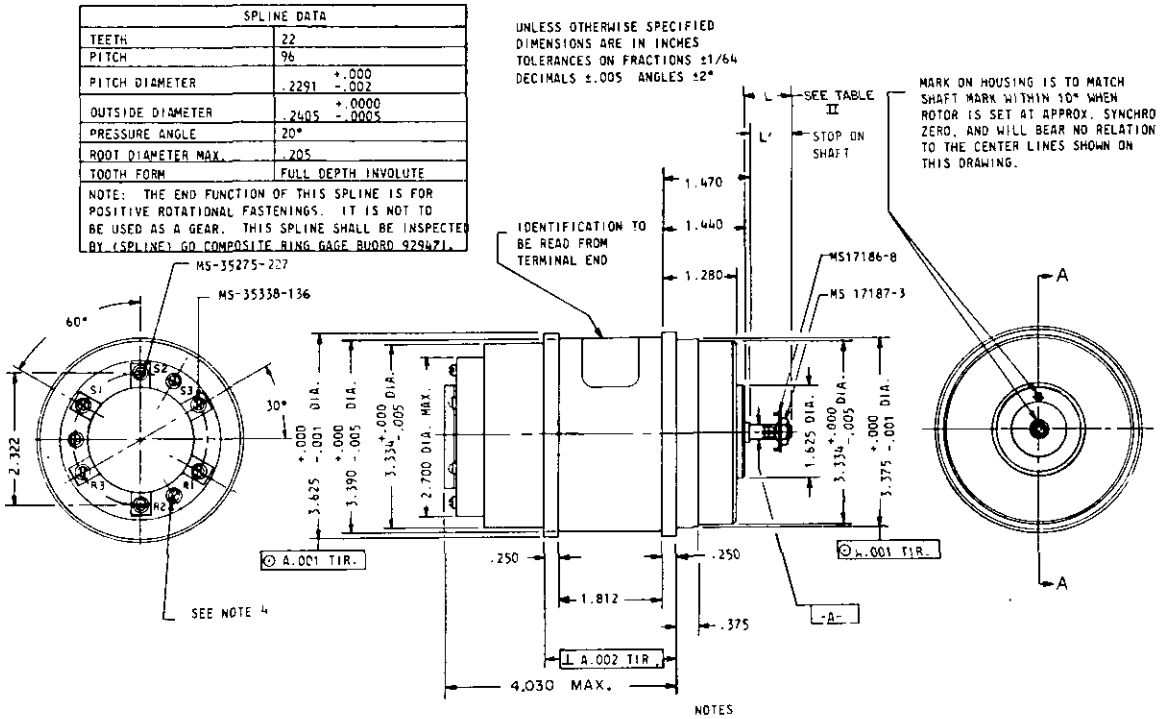
This Specification has been Approved by the Ordnance Systems Command, Department of the Navy

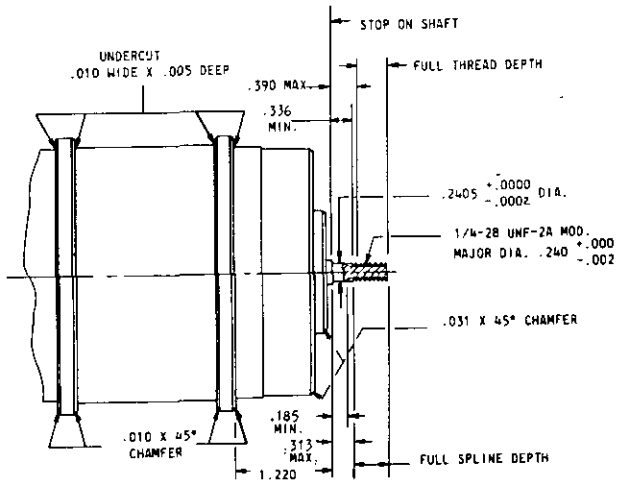
THE COMPLETE REQUIREMENTS FOR PROCURING THE SYNCHRO DESCRIBED HEREIN SHALL CONSIST OF THIS DOCUMENT AND THE ISSUE IN EFFECT OF SPECIFICATION MIL-S-20708.

SPLINE DATA	
TEETH	22
PITCH	96
PITCH DIAMETER	.2291 +.000 -.002
OUTSIDE DIAMETER	.2405 +.0000 -.0005
PRESSURE ANGLE	20°
ROOT DIAMETER MAX.	.205
TOOTH FORM	FULL DEPTH INVOLUTE

NOTE: THE END FUNCTION OF THIS SPLINE IS FOR POSITIVE ROTATIONAL FASTENINGS. IT IS NOT TO BE USED AS A GEAR. THIS SPLINE SHALL BE INSPECTED BY (SPLINE) GO COMPOSITE RING GAGE DWORD 929471.

UNLESS OTHERWISE SPECIFIED
DIMENSIONS ARE IN INCHES
TOLERANCES ON FRACTIONS ±1/64
DECIMALS ±.005 ANGLES ±2°





PARTIAL SECTION
A - A

NOTES

1. PERPENDICULARITY AND CONCENTRICITY SHALL BE MEASURED WITH THE UNIT IN A VERTICAL POSITION SUPPORTED BY THE SHAFT, THEN THE HOUSING ROTATED.
2. RUNOUT OF FEATURE "A" (SMOOTH PORTION) OF SHAFT SHALL NOT EXCEED .0008 TIR.
3. END PLAY AND RADIAL PLAY SHALL BE MEASURED DURING THE REVERSAL OF THE FOLLOWING LOADS:

   END PLAY - 1 POUND RADIAL PLAY - 1 POUND

4. THE TERMINAL BLOCK SHALL BE SECURED BY SCREWS OTHER THAN THE TERMINAL SCREWS. RECESSES SHALL ALLOW TERMINAL LUGS MS-25036-6 TO BE INSTALLED WITH THE BARRELS INBOARD OR OUTBOARD, AND PREVENT THE LUGS FROM TURNING.
5. MINOR VARIATIONS OF UNIT CONFIGURATION ARE PERMITTED FOR UNDIMENSIONED DESIGN DETAIL.
6. THE FOLLOWING LIST OF HARDWARE, MS-35275-222 MACHINE SCREW (6 REQUIRED), MS-35338-136 LOCK WASHER (6 REQUIRED), MS-17186-8 DRIVE WASHER (1 REQUIRED) MS-17187-3 NUT (1 REQUIRED), SHALL BE PACKAGED IN A VENTILATED ENVELOPE MADE OF MATERIAL HAVING A PH VALUE OF 6-8 THAT IS COMPATIBLE WITH THE WEIGHT OF THE CONTENTS. THEN THE PACKAGED HARDWARE SHALL BE PLACED IN THE UNIT SYNCHRO PACKAGE. FOR INFORMATION ONLY, SEE SYNCHRO CLAMP MS-17183.
7. PENDING ISSUANCE OF A QUALIFIED PRODUCTS LIST (QPL) A SAMPLE OF THE SYNCHROS TO BE FURNISHED UNDER THIS SPECIFICATION SHALL BE SUBJECTED TO THE PREPRODUCTION TESTS AS SPECIFIED IN MIL-S-20708.

FSC 5990

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TABLE I

Requirement	Value	Unit	Tolerance
Frequency	60	Hz	±1%
Primary Voltage	90	volts	nominal
Primary Current	375	milliamps	maximum
Primary Power	2.5	watts	nominal
Impedance:			
Zso	197-267	ohms	min.-max.
Zrs	24.2-32.7	ohms	min.-max.
Impedance Angle:			
Zso	82-86	degrees	min.-max.
Zrs	40-60	degrees	min.-max.
Transformation Ratio	1.154	-------	±2%
Phase Shift	2.0	degrees	±1.0
Electrical Error	8.0	minutes	maximum
Null Voltage:			
Total	100	millivolts	maximum
Fundamental	40.0	millivolts	maximum
Friction Torque	0.5	ounce-inches	maximum
Radial Play	0.0008	inches	maximum
End Play	0.0025	inches	±0.0015
Temperature Rise	10.0	degrees(C)	maximum

This unit shall not be subjected to the altitude requirements of MIL-S-20708.

Part number: M20708/75A plus applicable dash number from Table II.

TABLE II

DASH NO.	SHAFT DIAMETER	SHAFT STYLE	TERMINAL END	L' DIMENSION TO STOP ON SHAFT	L REFERENCE FREE SHAFT LENGTH
*-001	.2405	Splined	Terminal Board	.700 ± .010	.730 ± .010

*Dash number -001 shall be the NATO Standard.

Custodians: Navy-AS

Review Activities: Navy-SH,OS

User Activities: Marine Corp. MC; U.S. Coast Guard-CG

Preparing Activity: Navy-OS

Project No. 5990-N-041

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